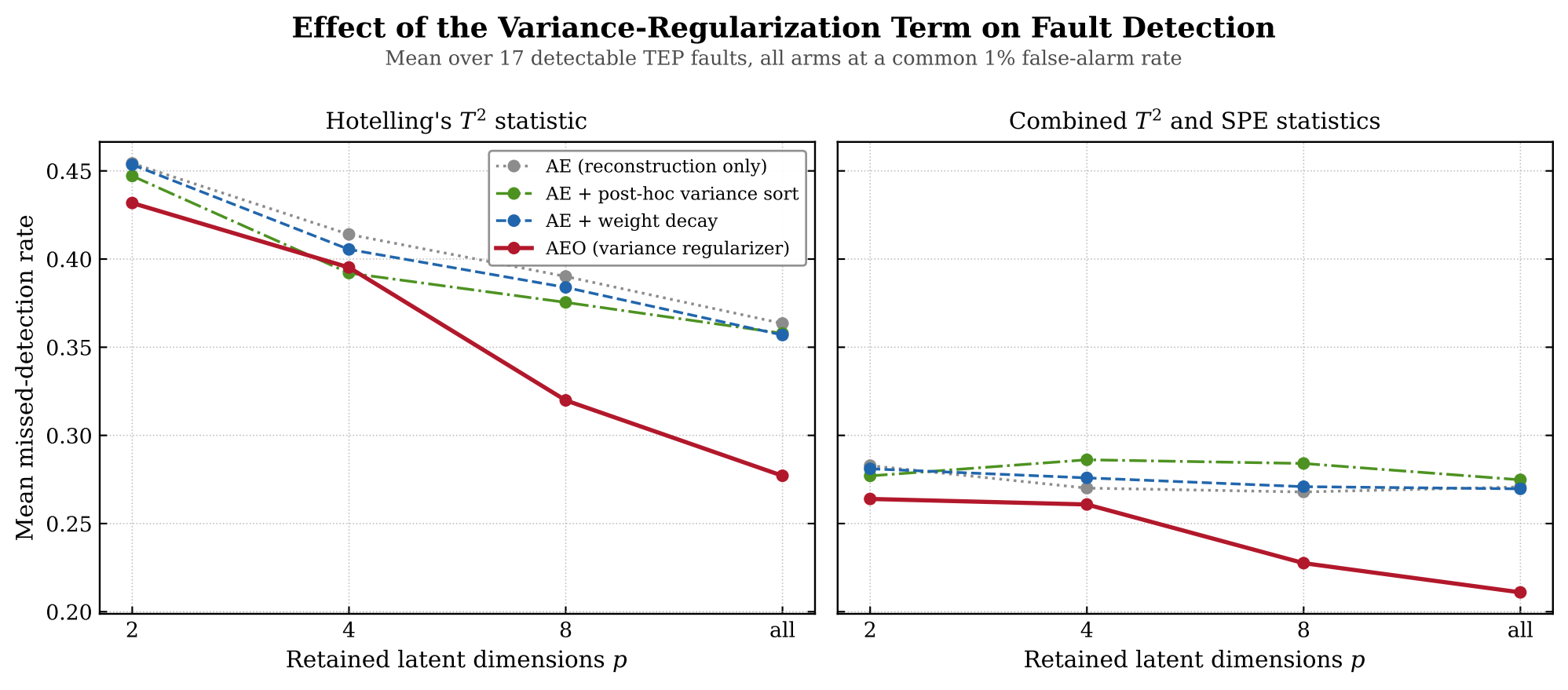
Effect of the Variance-Regularization Term on Fault Detection
Mean over 17 detectable TEP faults, all arms at a common 1% false-alarm rate
Hotelling's T^2 statistic
Combined T^2 and SPE statistics
Mean missed-detection rate
0.45
0.40
0.35
0.30
0.25
0.20
AE (reconstruction only)
AE + post-hoc variance sort
AE + weight decay
AEO (variance regularizer)
2
4
8
all
Retained latent dimensions p
Retained latent dimensions p